

CodeAlpha Internship Project Report

Project 2: FAQ Chatbot

Objective: Develop an FAQ chatbot that answers user questions by matching them with a predefined FAQ dataset using Natural Language Processing (NLP).

Tools & Technologies: Python, NLTK, scikit-learn, Flask/Streamlit, Pandas.

Features:

- FAQ dataset
- Text preprocessing (tokenization, stop-word removal)
- Cosine similarity for question matching
- Displays the best matching answer
- Simple chat interface

Algorithm:

1. Load FAQ dataset.
2. Preprocess user query.
3. Convert text using TF-IDF.
4. Calculate cosine similarity.
5. Return the most relevant answer.

Sample Python Code:

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
```

Expected Output:

User: What are your working hours?

Bot: We are available Monday to Friday, 9 AM to 6 PM.

Future Enhancements: Voice chatbot, multilingual support, integration with LLMs, database connectivity.

Conclusion: The FAQ chatbot demonstrates NLP-based intent matching to provide quick and accurate responses.